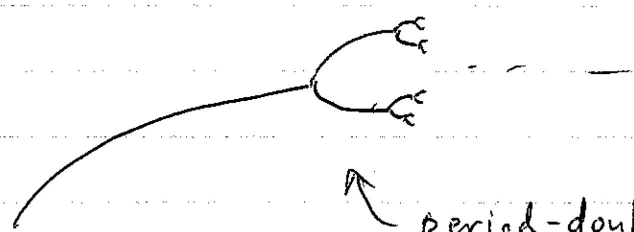
logistic map $\quad x_{n+1} = r x_n (1 - x_n)$, $\quad r \in [0, 4]$
$x \in [0, 1]$

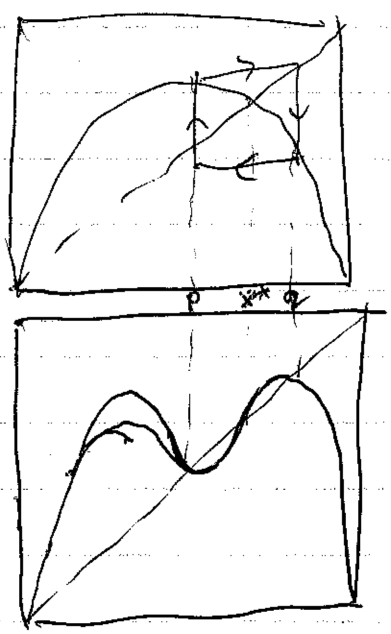
orbit diagram shows attractor as a function of r



period-doubling cascade

analyze 2-cycle by considering $f^2 = f \circ f$

f

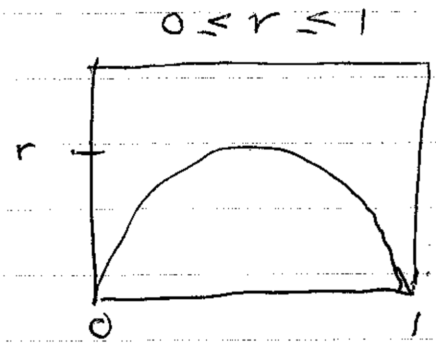


linear stability of 2-cycle:

$\lambda = f'(p) f'(q)$
$|\lambda| < 1 \Rightarrow$ stable

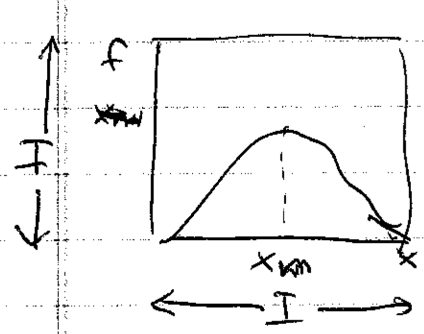
4-cycle : consider $f^4 = f \circ f \circ f \circ f$

$x_{n+1} = r\sin(\pi x_n)$

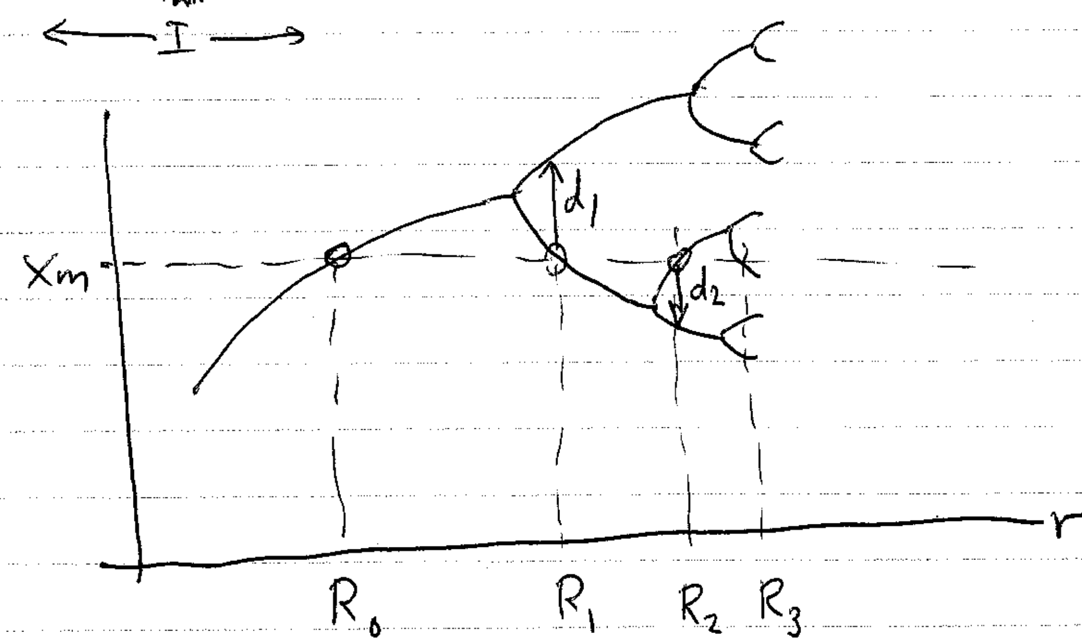


similar orbit diagram

Universal features of <u>unimodal maps</u>:



smooth, single max at x_m, concave down, $f''(x_m) < 0$, i.e. "quadratic max"



$$\lim_{n\to\infty} \frac{d_n}{d_{n+1}} = \alpha = -2.5029\ldots$$

What's special about R_j?

at ~~R_n~~ $r = R_{n+1}$, there is an n-cycle:

$$\{p_1, \dots, p_n\} \qquad f^n(p_j) = p_j \qquad j = 1, \dots n$$

$$\lambda \doteq \left.\frac{df^n}{dx}\right|_{x=p_1} = \frac{d}{dx}\Big(\underbrace{f(f(f \cdots f(x)))}_{n\text{-times}}\Big)$$

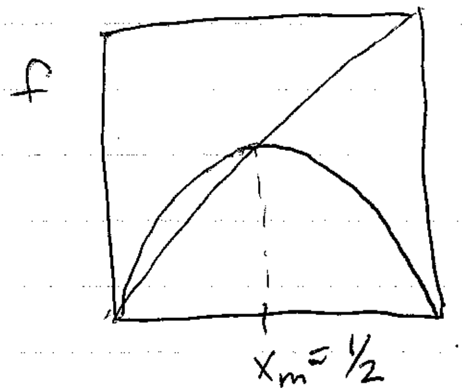
$$= f'(f^{n-1}(p_1))\, f'(f^{n-2}(p_1)) \cdots f'(p_1)$$

$$= f'(p_n)\, f'(p_{n-1}) \cdots f'(p_1)$$

at $r = R_{n+1}$, one of the p's is at x_m, so $\lambda = 0$

the n-cycle is "superstable"

Observation:

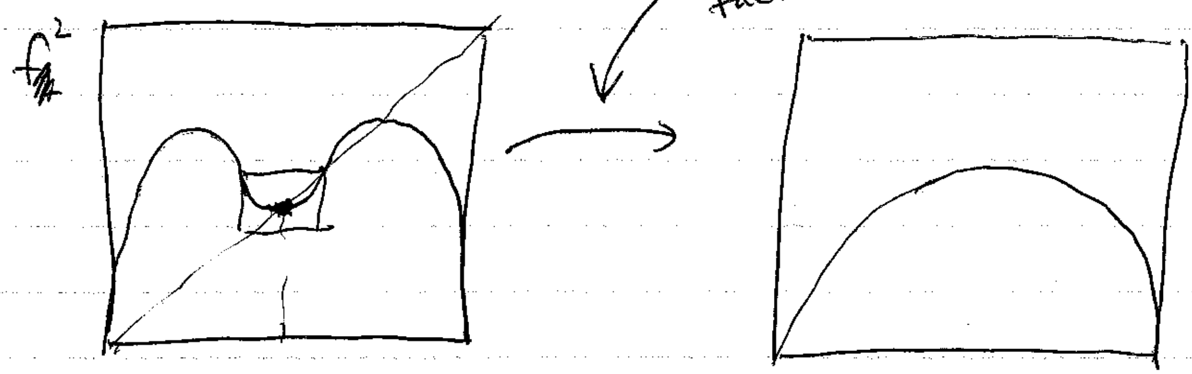


$r = R_0$

$$\frac{1}{2} = f\left(\frac{1}{2}, R_0\right)$$

$$\frac{1}{2} = R_0 \frac{1}{4} \Rightarrow R_0 = 2$$

use scale factor α



$r = R_1$

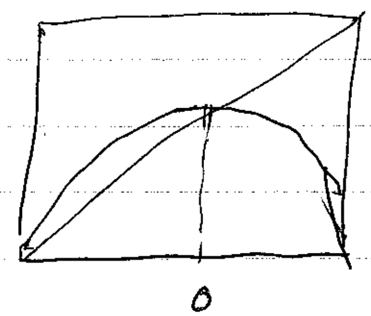
$$\frac{1}{2} = f^2\left(\frac{1}{2}, R_1\right)$$

To simplify the analysis, shift x ~~xm~~ so that $x_m = 0$.

(For logistic map, let $y = x - \frac{1}{2}$)

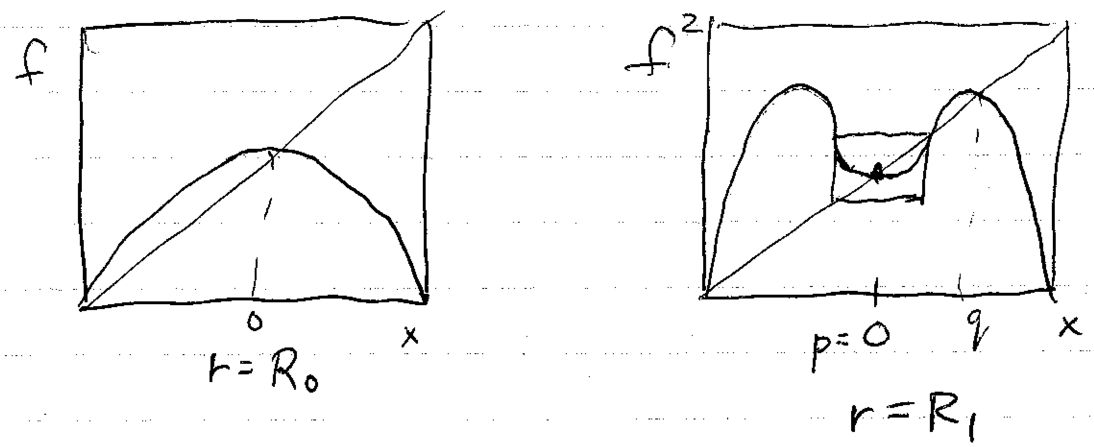
$$\boxed{\begin{aligned} y_{n+1} &= x_{n+1} - \frac{1}{2} \\ &= r x_n (1 - x_n) - \frac{1}{2} \\ &= r\left(y_n + \frac{1}{2}\right)\left(1 - y_n - \frac{1}{2}\right) - \frac{1}{2} \\ &= r\left(\frac{1}{4} - y_n^2\right) - \frac{1}{2} \end{aligned}}$$

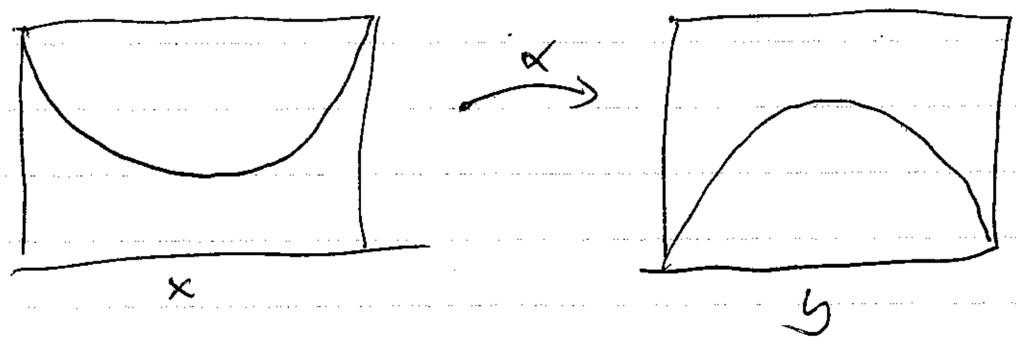
$x_{n+1} = f(x_n, r)$



at $r = R_0$, $x^* = 0$ is a superstable fixed-pt

at $r = R_1$, 2 cycle ~~where~~ $\{p, q\}$, where $p = 0$ so 2-cycle is super stable.





$$y_n = \alpha x_n$$
$$y_{n+1} = \alpha x_{n+1} = \alpha f^2(x_n) = \alpha f^2\left(\frac{y_n}{\alpha}\right)$$

Claim: $f(x, R_0) \approx \alpha f^2\left(\frac{x}{\alpha}, R_1\right)$

$f^2(x, R_1) \approx \alpha f^4\left(\frac{x}{\alpha}, R_2\right)$, etc.

$$f(x, R_0) \approx \alpha f^2\left(\frac{x}{\alpha}, R_1\right) \approx \alpha^2 f^4\left(\frac{x}{\alpha^2}, R_2\right) \approx \cdots$$

Feigenbaum found (numerically)

$$\lim_{n\to\infty} \alpha^n f^{(2^n)}\left(\frac{x}{\alpha^n}, R_n\right) = g_0(x)$$

exists provided $\alpha = -2.5029$

and $g_0(x)$ is same for any unimodal function $f(x)$

Perhaps not so surprising since procedure involves zooming in on X_m

Can actually construct a sequence of functions; $g_0(x), g_1(x), g_2(x), \ldots$ with different starting pts.

$$g_1(x) = \lim_{n\to\infty} \alpha^n f^{(2^n)}\left(\frac{x}{\alpha^n}, R_{n+1}\right)$$

$$g_2(x) = \ldots$$

$g_\infty(x)$

for $g_\infty(x)$: $\quad g_\infty(x) = \alpha g^2\left(\frac{x}{\alpha}\right)$; dropping R shifts

$$g(x) = \alpha g^2\left(\frac{x}{\alpha}\right)$$

augment with two conditions

$g'(0) = 0$ $x_m = 0$ w.l.o.g.

$g(0) = 1$ sets scale for x w.l.o.g.

$$\Rightarrow \quad g(0) = 1 = \alpha g^2(0) = \alpha g(1) \quad \Rightarrow \quad \alpha = \frac{1}{g(1)}$$

$$g(x) = 1 + c_2 x^2 + c_4 x^4 + \dots = \alpha g^2\left(\frac{x}{\alpha}\right)$$

$$= \alpha g\left(1 + c_2\left(\frac{x}{\alpha}\right)^2 + c_4\left(\frac{x}{\alpha}\right)^4 + \dots\right)$$

$$= \alpha\left(1 + c_2\left(1 + c_2\left(\frac{x}{\alpha}\right)^2 + \dots\right)^2 + \dots\right)$$

3 terms:

$$\left.\begin{aligned} 1 &= \alpha + \alpha c_2 + \alpha c_4 \\ c_2 &= \dots \\ c_4 &= \dots \end{aligned}\right\}$$

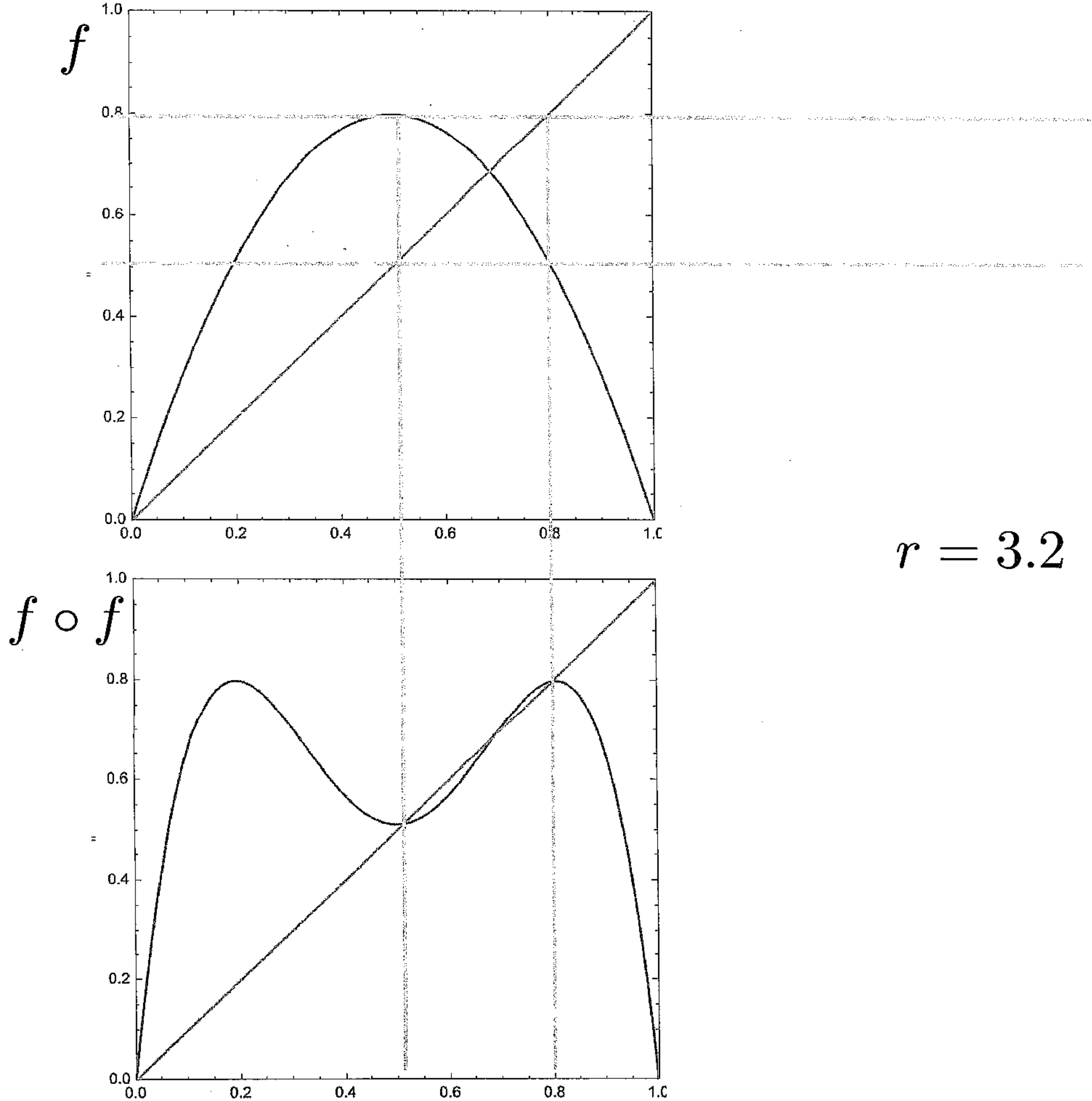
$$\left.\begin{aligned} \alpha &= -2.53403\dots \\ c_2 &= -1.52224\dots \\ c_4 &= 0.127 \end{aligned}\right\} \text{3 terms}$$

vs.

$$\left.\begin{aligned} \alpha &= -2.5029\dots \\ c_2 &= -1.5276\dots \\ c_4 &= 0.1048\dots \end{aligned}\right\} \text{Feigenbaum's 7 terms}$$

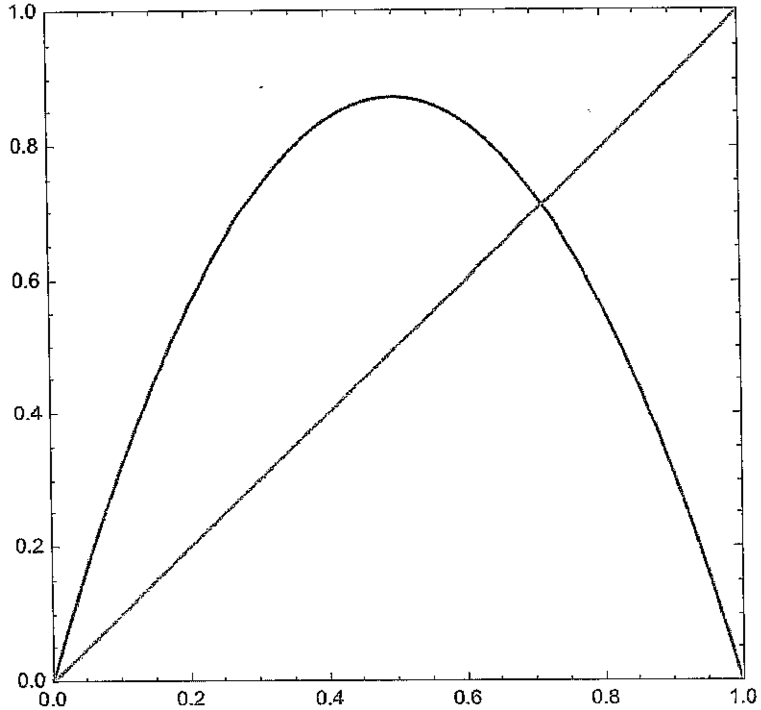
2^{nd} Feigenbaum constant

$$\delta = \lim_{n\to\infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} = 4.669\ldots$$



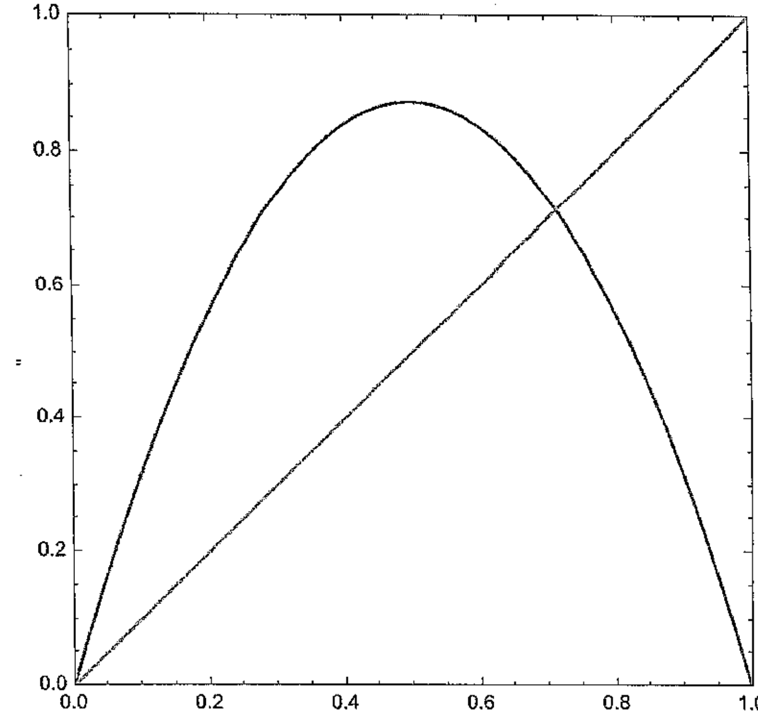
$r = 3.2$

f

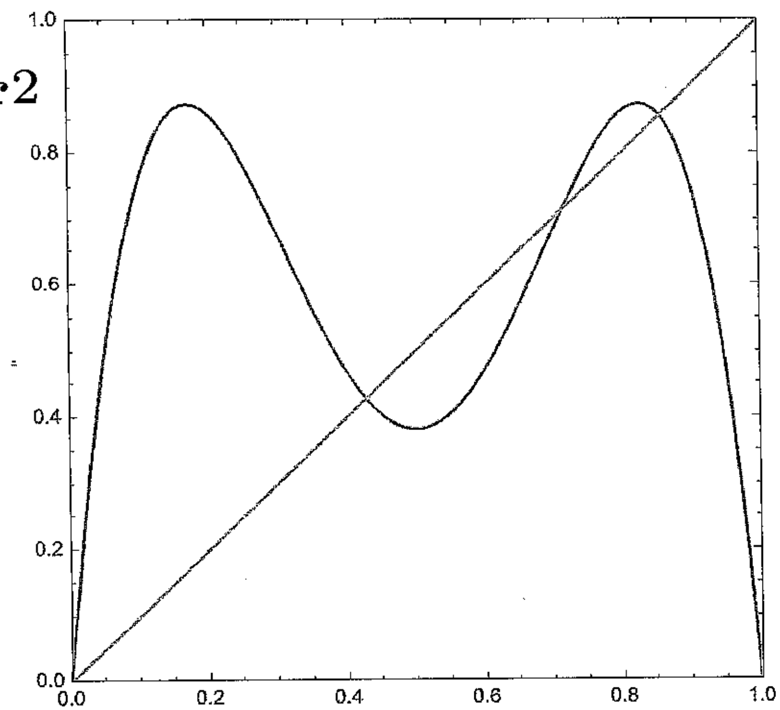


$r = 3.5$

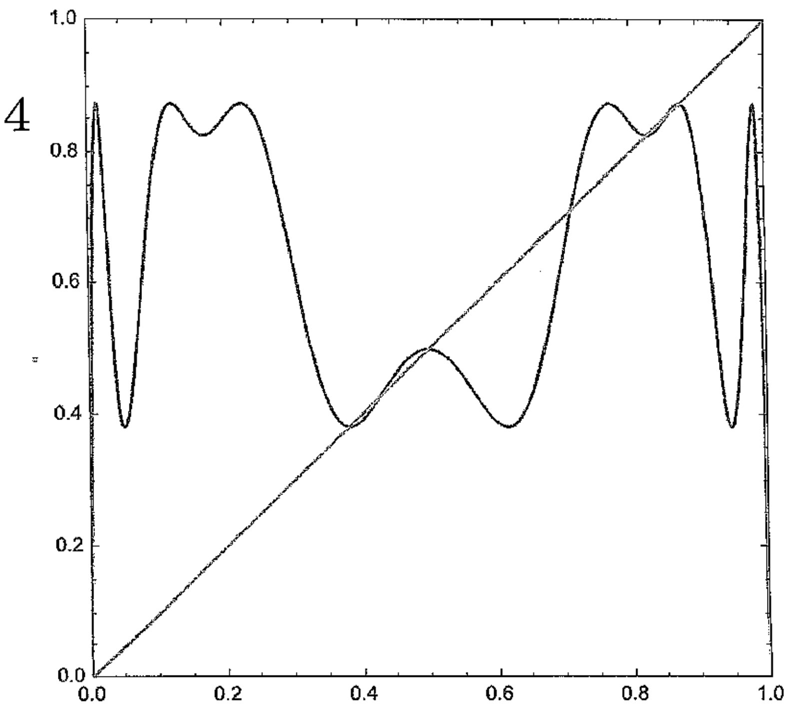
f



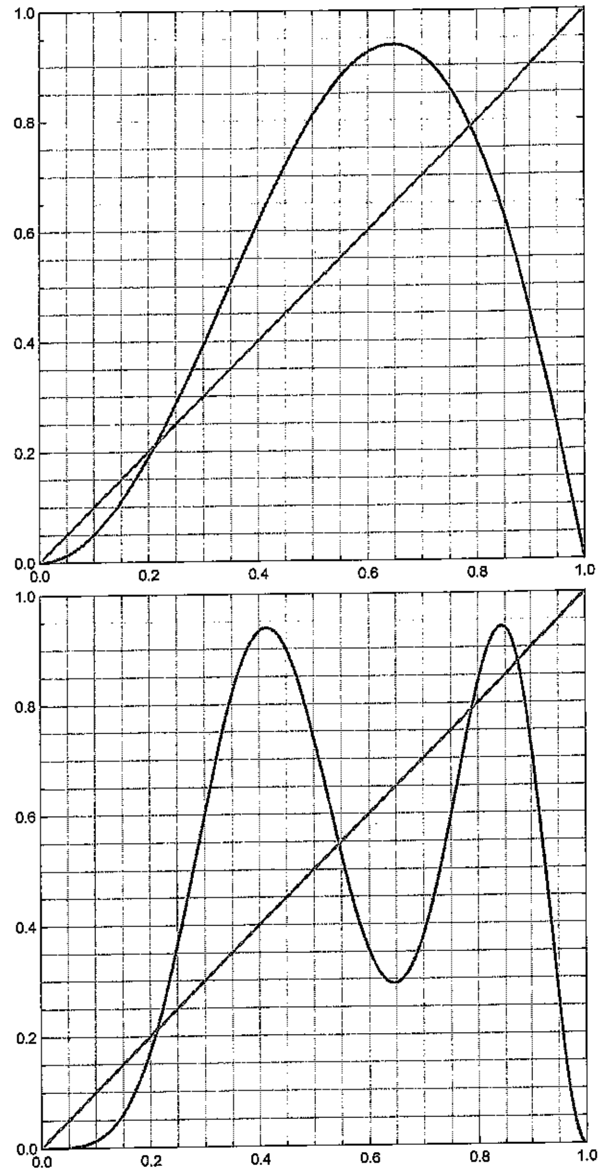
f^2



f^4



2. (15 points) Consider the unimodal map $x_{n+1} = f(x_n)$ (top plot below) and its second iterate $f^2(x_n) \equiv f(f(x_n))$ (bottom plot below).



(a) (4 points) Use this graphical information to determine the approximate coordinates of a 2-cycle, which you can label with x-values p, q in the bottom plot. (For each map the diagonal $x_{n+1} = x_n$ is also plotted.)

(b) (6 points) As accurately as possible plot the cobweb diagram associated with the 2-cycle on the top plot, i.e. the cobweb diagram associated with an initial condition $x_0 = p$.

(c) (5 points) Is the 2-cycle stable or unstable? Justify your answer.